

P2 — Backend + Database

SIH 26143 · Detailed 6-Phase Work Plan · 6 owned work packages

ROLE OBJECTIVE

Own the system backbone: persistence, APIs, orchestration, provenance, security and reliable service-to-service interfaces.

HOW TO USE THIS PLAN

Each package has four parts: what to build/do, the concrete deliverable, dependencies, and an acceptance condition. The owner is responsible for making the output usable by the next phase—not merely completing a task locally.

PHASE 1 — DATA & PREPROCESSING

Work package 1: Data storage architecture

Design PostgreSQL/PostGIS entities for scenes, detections, spill geometries, incidents, source corridors, AIS observations, candidate vessels, evidence features and provenance; define primary/foreign keys, spatial indexes and timestamps.

Deliverable: schema/ERD + migrations + indexing plan.

Dependencies / acceptance: Depends on: data contracts from P1/P4/P5. Acceptance: all core outputs can be persisted with provenance.

PHASE 2 — SPILL DETECTION

Work package 2: Detection API + persistence

Build the API endpoint/workflow for SAR scene submission and detection retrieval; validate inputs; store mask metadata, geometry, confidence and processing status; expose incident IDs for downstream services.

Deliverable: working detection API + persistence layer + API examples.

Dependencies / acceptance: Depends on: P1 inference interface. Acceptance: a scene can move from request to stored detection result.

PHASE 3 — SOURCE RECONSTRUCTION

Work package 3: Reconstruction service

Implement the service contract between detection, environmental forcing and drift modelling; create job/status handling, result persistence and failure states; store source corridor, origin window and uncertainty metadata.

Deliverable: reconstruction API/service workflow.

Dependencies / acceptance: Depends on: P5 OpenDrift workflow and P1 feature contract. Acceptance: successful and failed reconstruction jobs are both represented explicitly.

PHASE 4 — AIS CORRELATION

Work package 4: AIS data service

Create AIS ingestion/query endpoints; validate timestamps and coordinates; index vessel observations by time and geography; support queries around a reconstructed source corridor and origin window.

Deliverable: AIS storage + query API + indexed trajectory retrieval.

Dependencies / acceptance: Depends on: P4 AIS schema. Acceptance: API returns only observations matching requested spatial/temporal bounds.

PHASE 5 — EVIDENCE FUSION + UI

Work package 5: Evidence-fusion backend

Create models for evidence families, candidate vessels, feature values, evidence provenance, uncertainty, ranking state and abstention; implement deterministic scoring/fusion configuration rather than hiding logic in UI code.

Deliverable: evidence-fusion service + auditable stored results.

Dependencies / acceptance: Depends on: P4 feature definitions and P5 uncertainty rules. Acceptance: candidate result can be traced back to component evidence.

PHASE 6 — VALIDATION & DEPLOYMENT

Work package 6: Backend integration + security

Connect all service endpoints; add authentication/authorization, input validation, error handling, logging, database constraints and audit events; test invalid inputs and partial service failures.

Deliverable: integrated backend + security checklist + API test suite.

Dependencies / acceptance: Depends on all upstream services. Acceptance: core workflow runs end-to-end and failures return explicit states instead of fabricated outputs.

CROSS-TEAM HANDOFF RULE

No component is considered complete if its output cannot be consumed by the next owner. Preserve IDs, timestamps, coordinate systems, provenance, uncertainty and failure states across every handoff.

Core project rule: AIS gaps, uncertain metocean forcing, ambiguous SAR signatures or weak evidence must increase uncertainty or trigger abstention—not be converted into fabricated confidence.